



Case Study: Security Incident Response

Ransomware Incident Analysis

Scenario:

A medium-sized healthcare provider has been hit with ransomware. Patient records are encrypted, and systems are offline.

Initial Assessment:

- Ransomware detected at 3:00 AM
- 50% of servers affected
- Electronic Health Records inaccessible
- Backup systems potentially compromised

Your Task:

- Develop incident response plan
- Identify containment strategies
- Plan recovery steps
- Document lessons learned



Practical Exercise: Vulnerability Assessment

Web Application Security Testing

Lab Environment:

You'll be working with a deliberately vulnerable web application in an isolated environment.

Testing Objectives:

- Identify SQL injection vulnerabilities
- Test for XSS weaknesses
- Assess authentication mechanisms
- Check for sensitive data exposure

Deliverables:

- Detailed vulnerability report
- Risk assessment matrix
- Remediation recommendations
- Prevention strategies